

1. Name four points in the diagram.

H, G, F, L

2. Name three collinear points in the diagram.

H, G, F

3. Name two lines in the diagram.

$\overleftrightarrow{LB}, \overleftrightarrow{HF}$

4. Name three coplanar points in the diagram.

C, K, E

5. Name the plane that is on the left in the diagram.

Plane HLF

6. Name the plane that is on the right in the diagram.

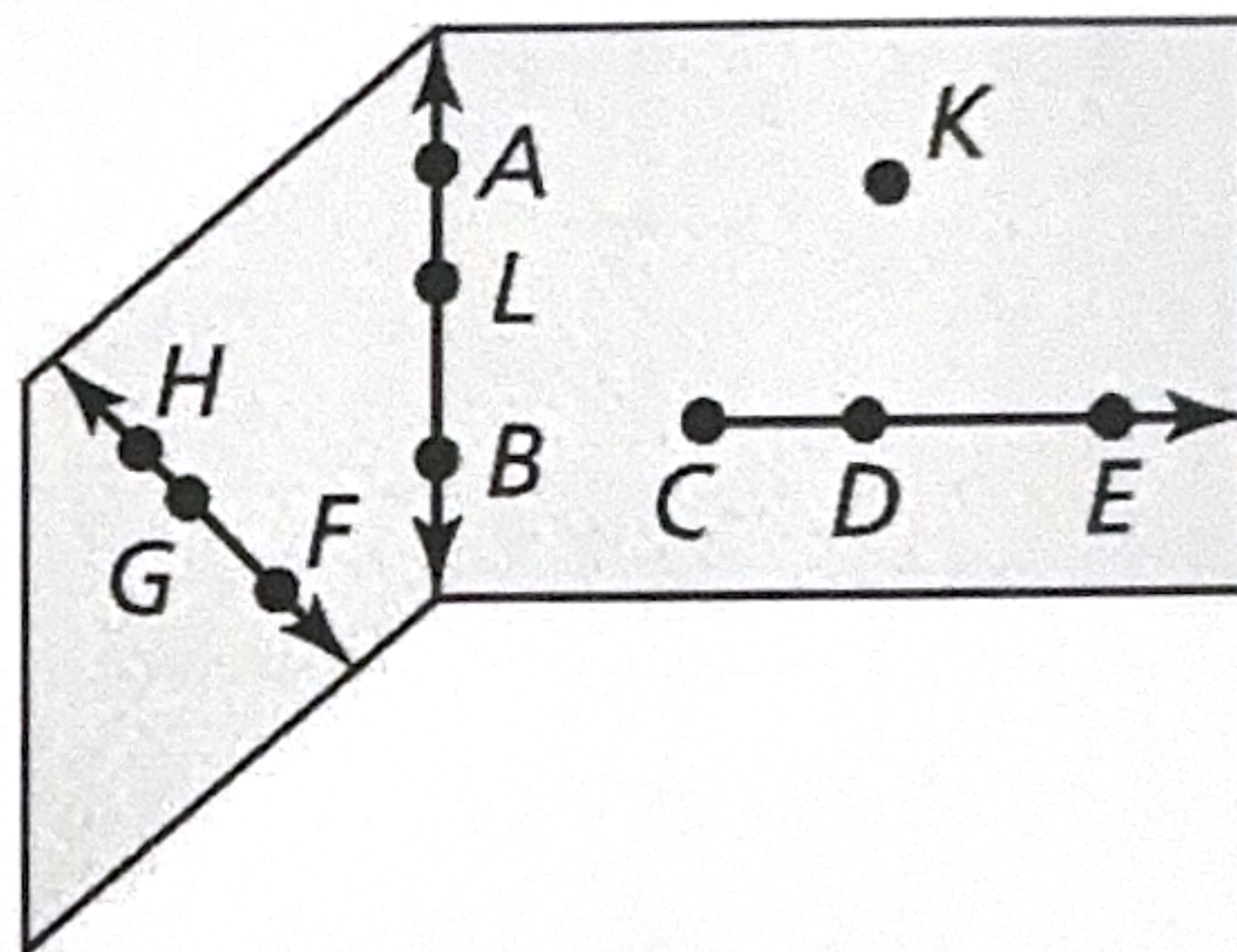
Plane CKE

7. Name three line segments in the diagram.

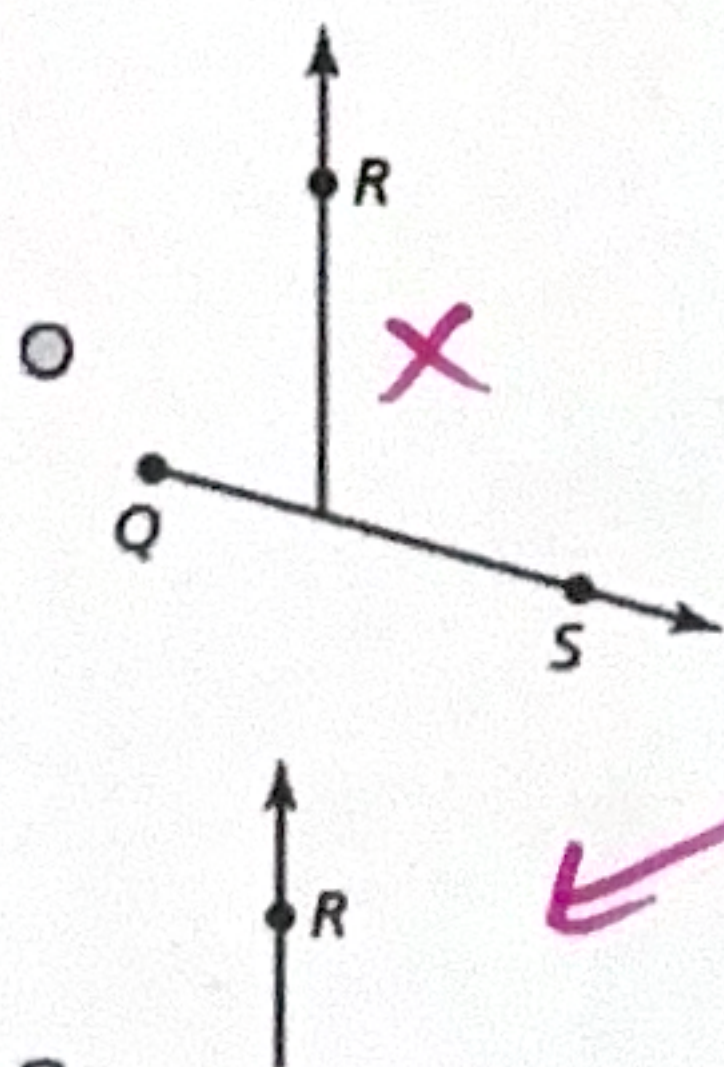
$\overline{GF}, \overline{LB}, \overline{CD}$

8. Name three rays in the diagram.

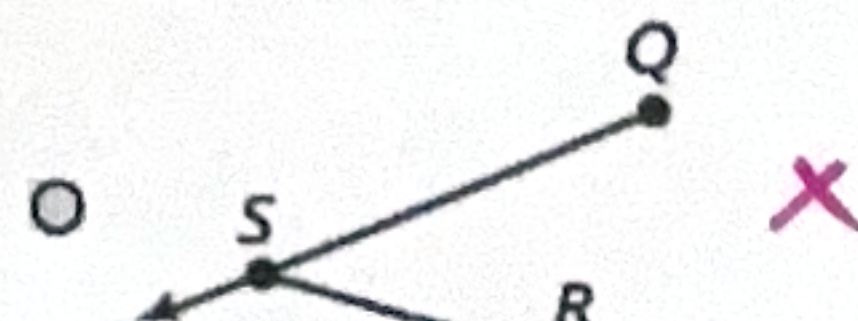
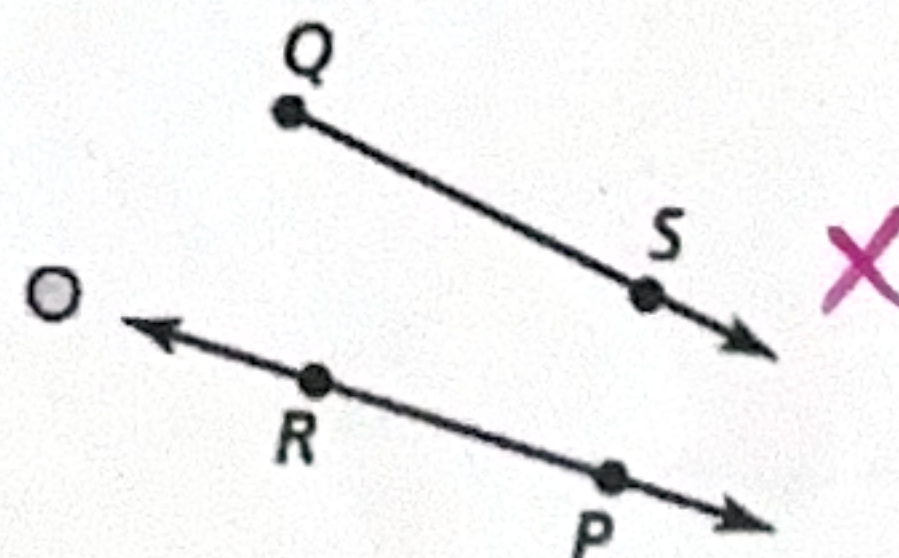
$\overrightarrow{CE}, \overrightarrow{LB}, \overrightarrow{GF}$



9. Which sketch shows $\overrightarrow{QR}$ and $\overleftrightarrow{QS}$?

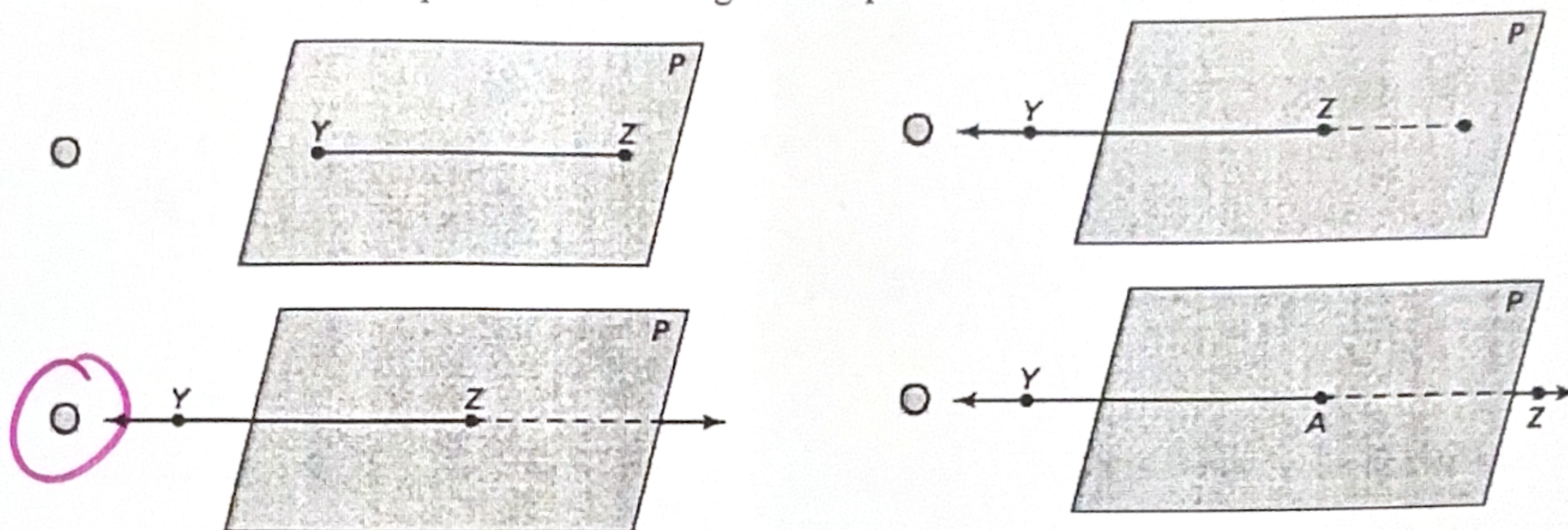


Ah!



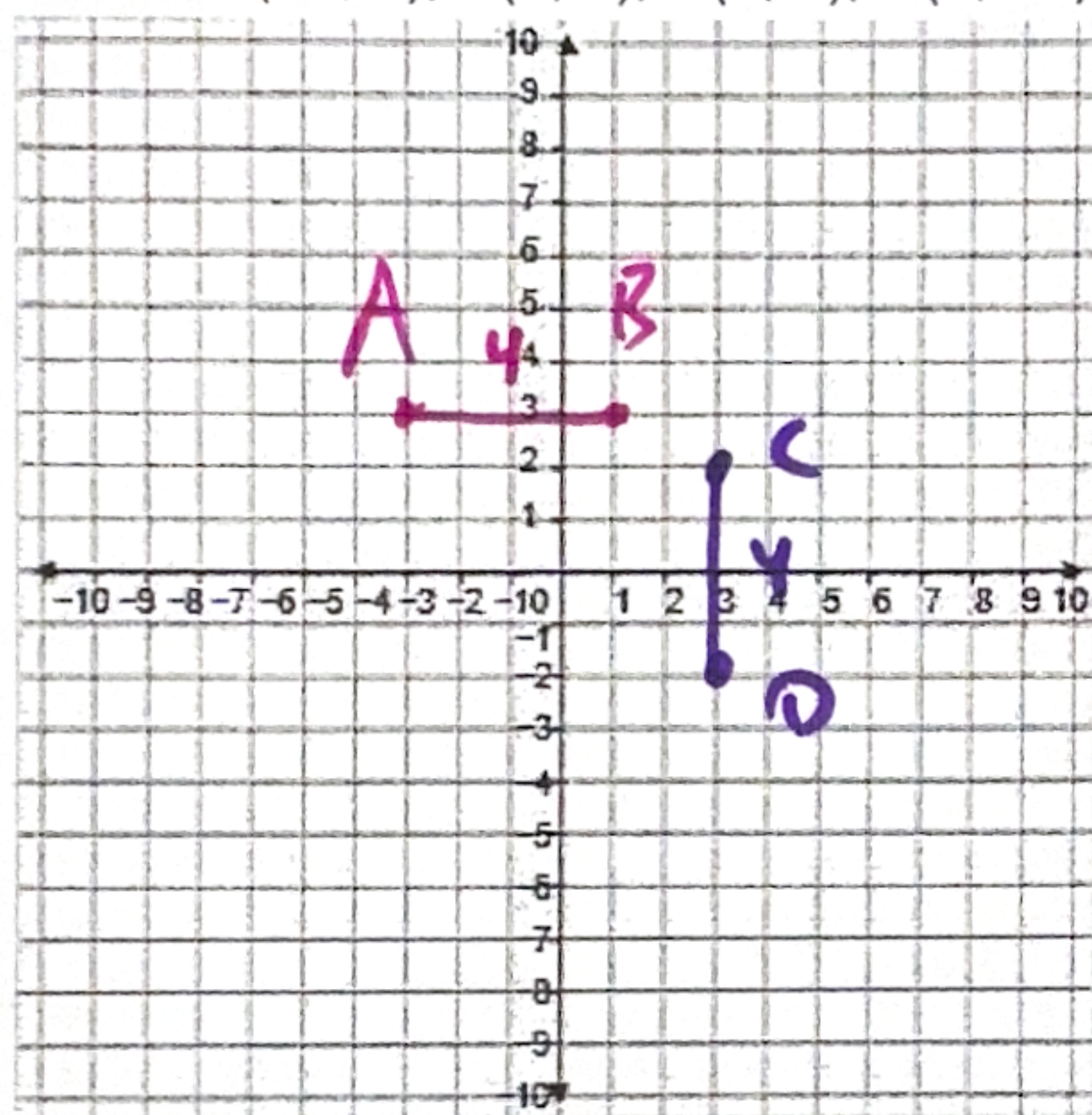
must be this one

10. Which sketch shows plane P intersecting $\overleftrightarrow{YZ}$ at point Z?



11. Plot the points:

$A(-3, 3)$, $B(1, 3)$, $C(3, 2)$, $D(3, -2)$



Are AB and CD congruent? Yes!
 $AB = CD = 4$

13. Find AC.



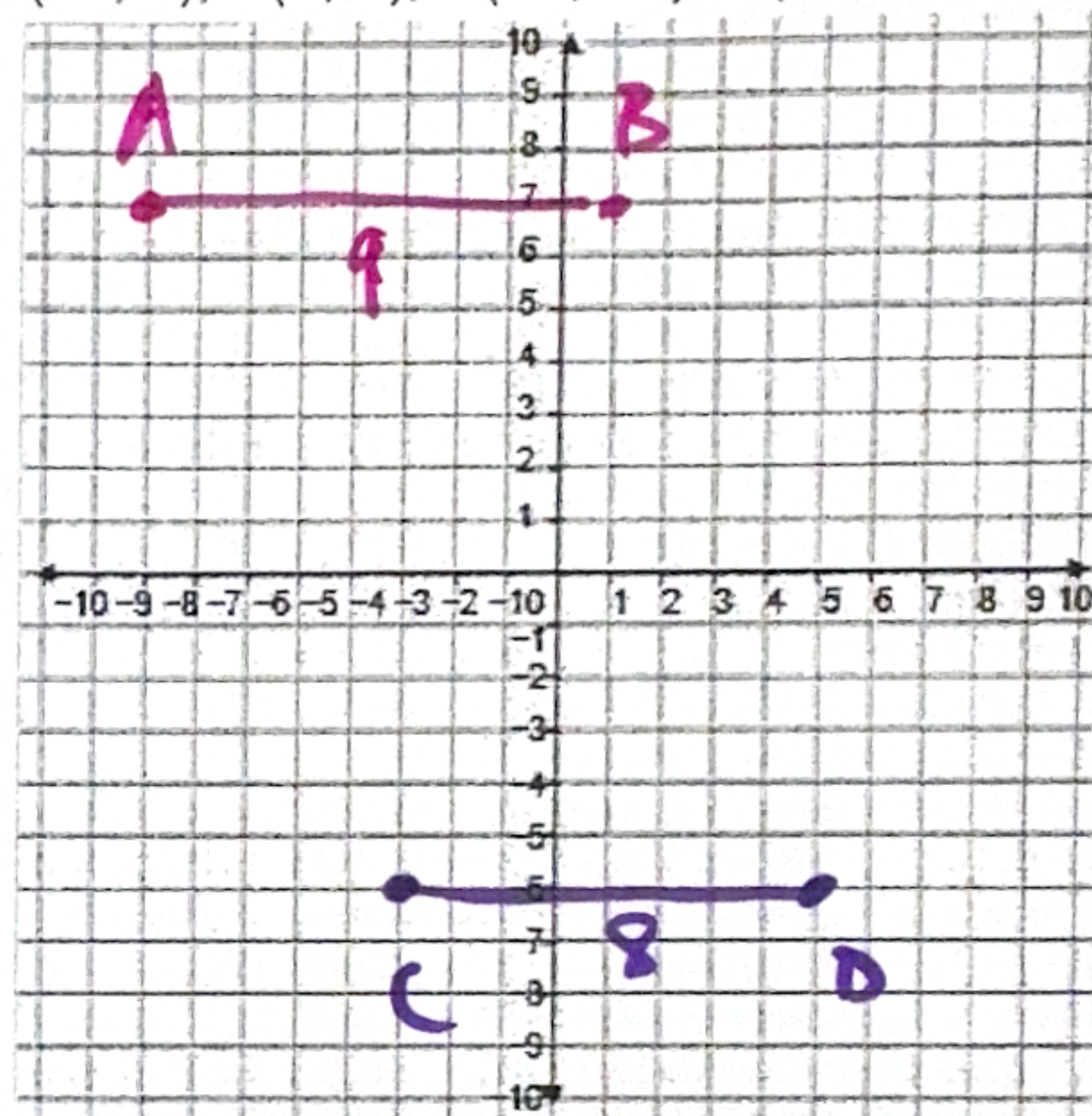
$$AC = 13 + 26$$

$$= 39$$

AC = 39

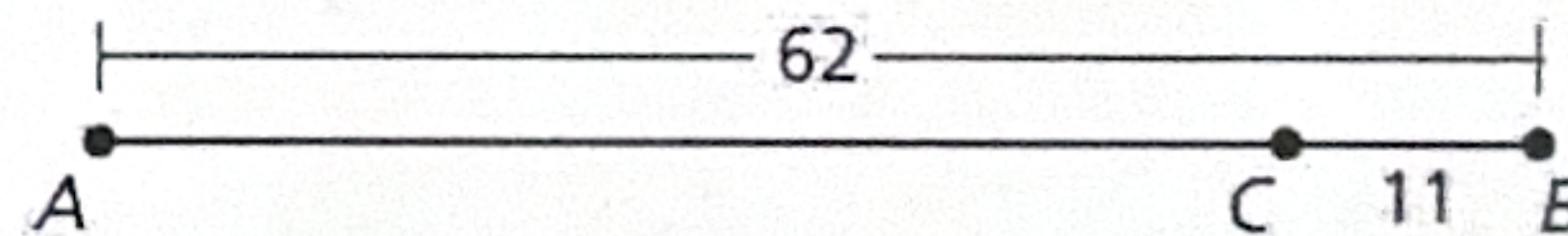
12. Plot the points:

$A(-8, 7)$, $B(1, 7)$, $C(-3, -6)$, $D(5, -6)$



Are AB and CD congruent? No!
 $AB = 9 \neq 8 = CD$

14. Find AC.



$$AB = AC + CB$$

$$62 = AC + 11$$

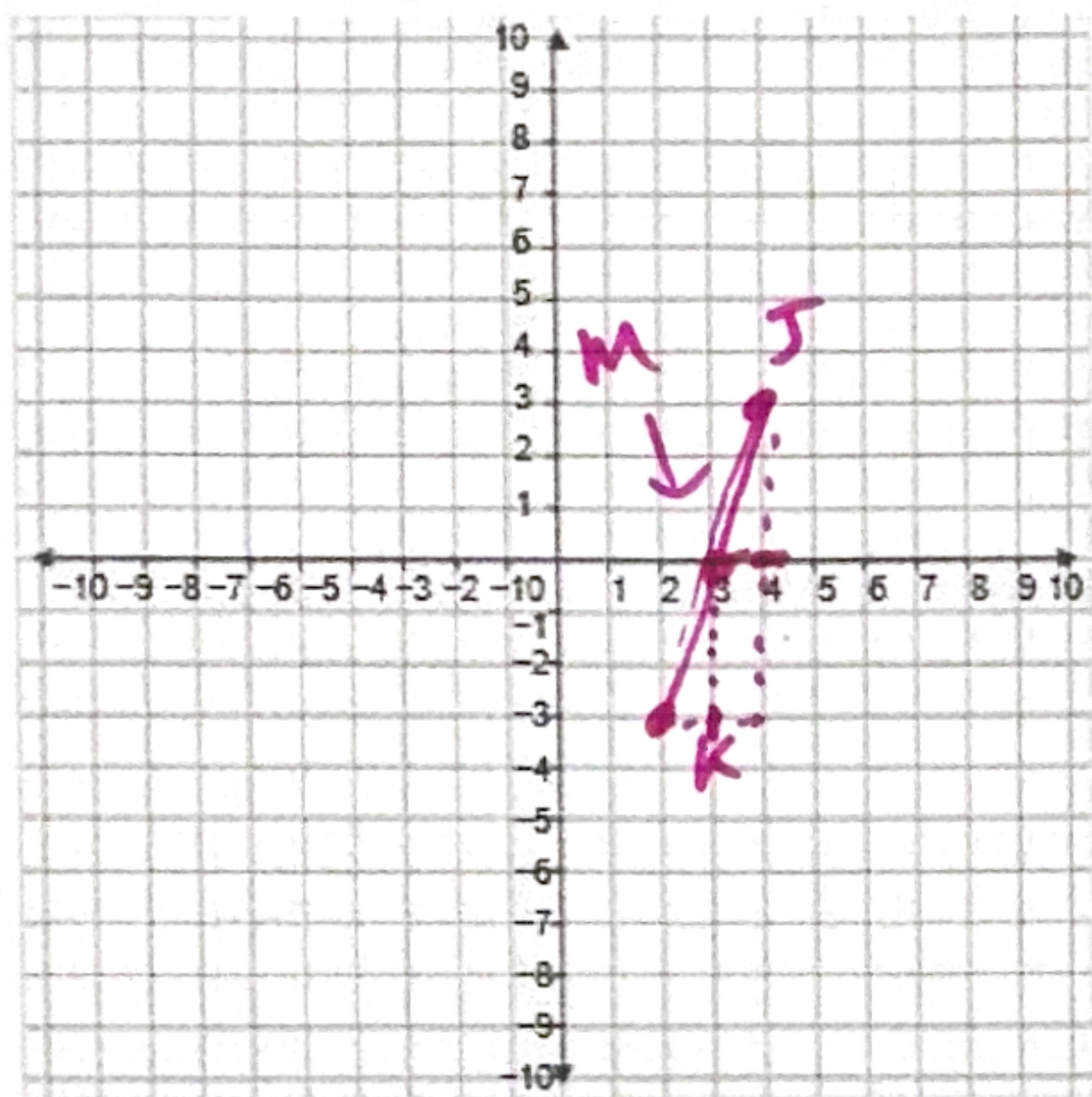
$$-11 \quad -11$$

AC = 51

15. Find the coordinates of midpoint M and the distance between J (4, 3) and K(2, -3). Round to the nearest tenth, if necessary.

Midpoint: M (3 , 0)

Length of JK ≈ 6.3 units



JK:

$$a^2 + b^2 = c^2$$

$$2^2 + 6^2 = c^2$$

$$4 + 36 = c^2$$

$$\sqrt{40} = \sqrt{c^2}$$

$$c = \sqrt{40} \approx 6.3 \text{ units}$$

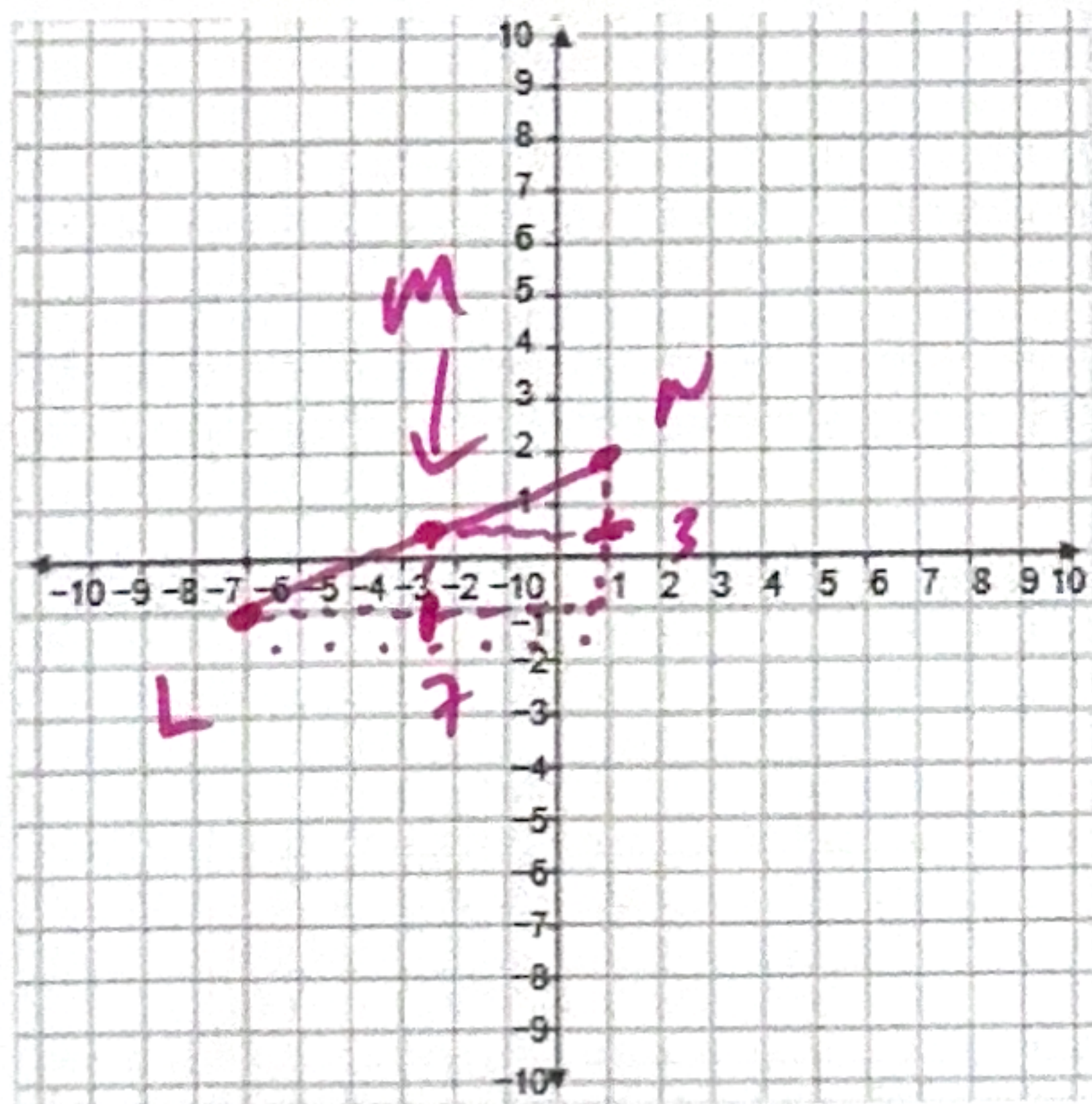
Note: if it asked for "simplest radical form"

$$\frac{\sqrt{40}}{\sqrt{4}\sqrt{10}} \rightarrow \boxed{2\sqrt{10}}$$

16. Find the coordinates of midpoint M and the distance between L(-6, -1) and N(1, 2). Express your answer in simplest radical form.

Midpoint: M (-2.5 , 0.5)

Length of LN = $\sqrt{58}$



LN:

$$a^2 + b^2 = c^2$$

$$7^2 + 3^2 = c^2$$

$$49 + 9 = c^2$$

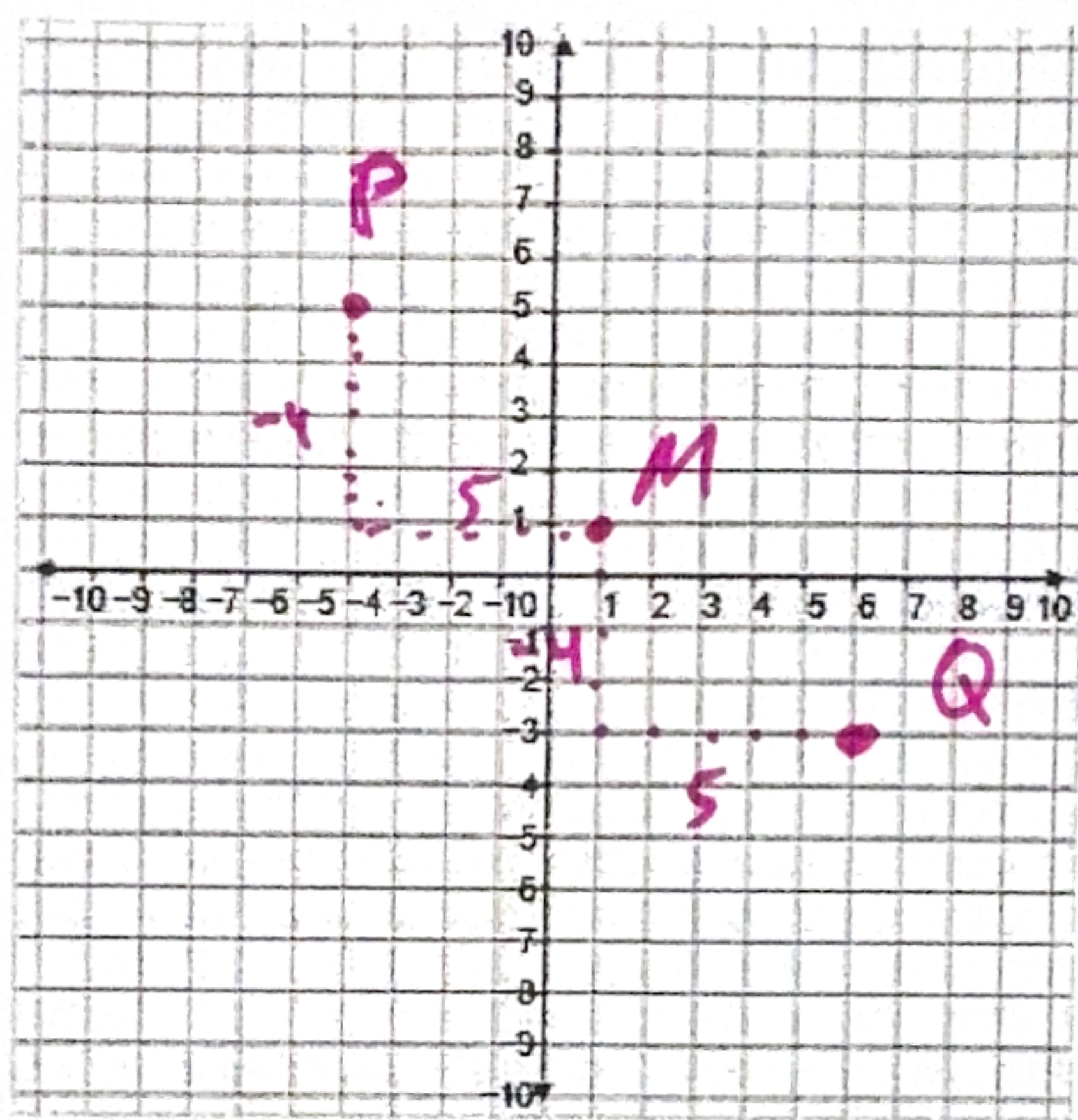
$$\sqrt{58} = \sqrt{c^2}$$

$$c = \sqrt{58}$$

$\nwarrow$ already in simplest radical form

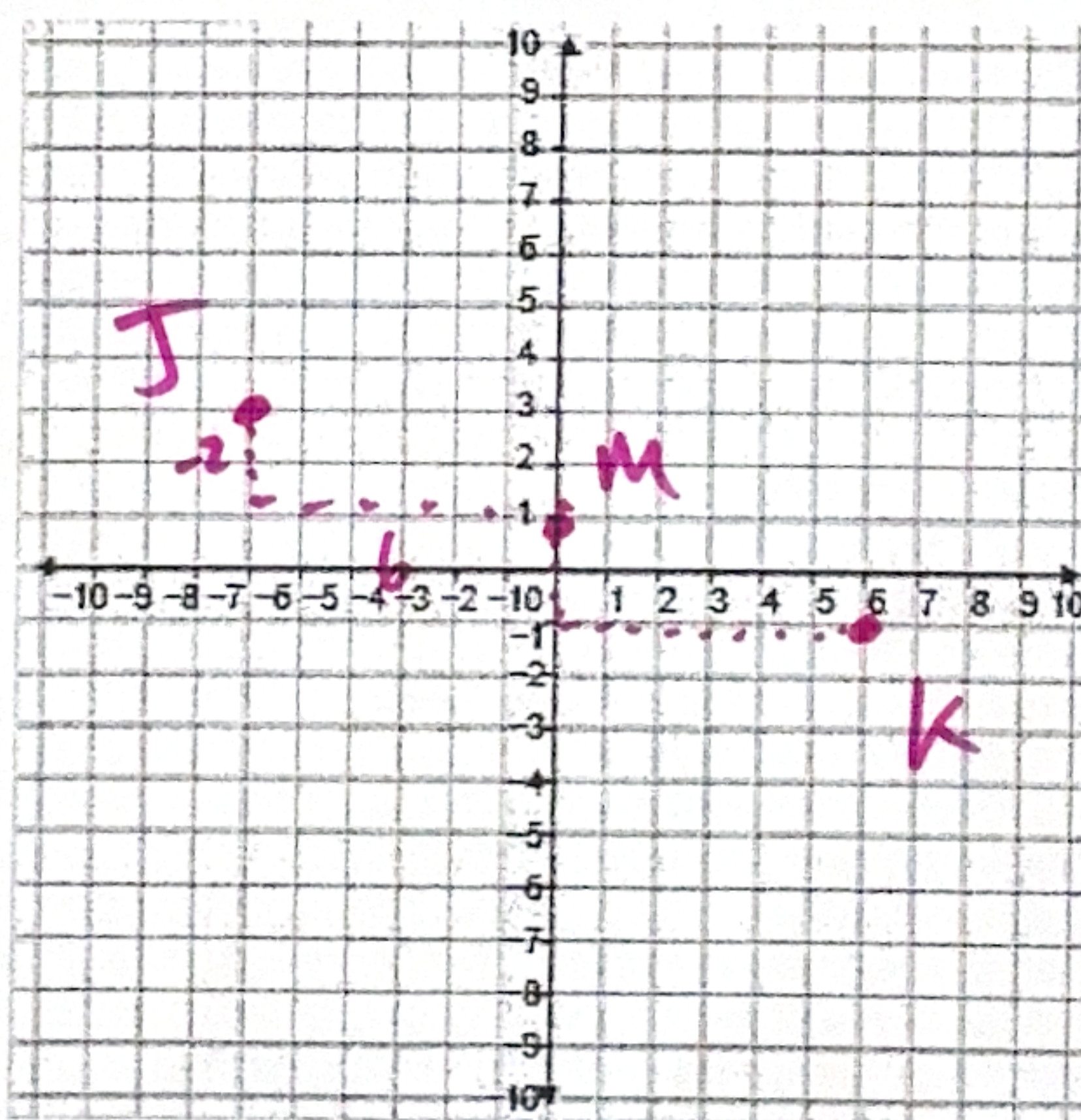
17. The midpoint of PQ is $M(1, 1)$. One endpoint is $P(-4, 5)$. Find the coordinates of endpoint Q .

$Q(6, -3)$

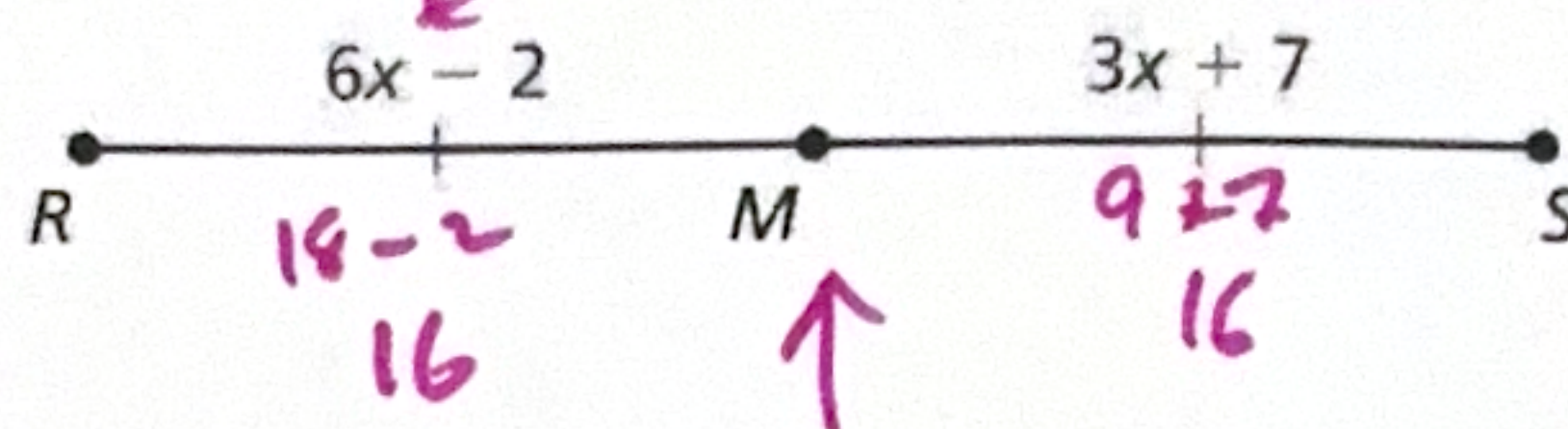


18. The midpoint of JK is $M(0, 1)$. One endpoint is $J(-6, 3)$. Find the coordinate of endpoint K .

$K(6, -1)$



19. Identify the segment bisector of RS. Then find RS.



$$6x - 2 = 3x + 7$$

$$-3x \quad -3x$$

$$3x - 2 = 7$$

$$+2 \quad +2$$

$$3x = 9$$

$$\frac{3x}{3} = \frac{9}{3}$$

$$x = 3$$

$$RS = 16 + 16$$

$$= 32$$

RS =

32

STOP – Check your answers against the answer key.

Optional Challenge

20. A circle is defined as “the set of all points that are the same distance from a center point.”

A circle whose center is the origin passes through the point $(-5, 12)$.

- a. Which of the following points also lies on this circle? Justify your answer.

(1) $(10, 3)$

(2) $(-12, 13)$

(3) $(11, 2\sqrt{12})$

(4) $(-8, 5\sqrt{21})$

- b. Determine the exact coordinates of four other points that also must lie on the same circle.

- c. Extra challenge: Can you determine the exact coordinates of an additional 3 points that you haven't used up till now?